

Juncheng Zhu

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EDUCATION

University of Waterloo

Bachelor of Computer Science

– René Descartes National Scholarship (\$25,000).

Waterloo, ON

Sept. 2025 – Present

HONOURS & AWARDS

USACO Gold Division: Algorithm Design | **CMO Qualifier:** Top 50 in Canada

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, SQL, Rust

Backend & Cloud: FastAPI, Flask, Nginx, AWS (EC2), Docker, Linux, Redis, Server-Sent Events (SSE)

Frontend: HTML/CSS, React, Chart.js

AI/ML & Data: PyTorch, RAG, pgvector, NumPy, Pandas

Tools & Databases: Git, GitHub Actions (CI/CD), Pytest, PostgreSQL, SQLite

PROJECTS

Dreamory: Memory Infrastructure for AI Companions | *Python, FastAPI, PostgreSQL* May 2026 – Present

- Architected a **three-tier hierarchical memory system** for companion LLMs (in-context state, time-decay hot zone, PostgreSQL cold storage) with **dual-axis (content + emotion) vector retrieval** via **pgvector** and BGE-M3 embeddings, enabling open-loop recall of months-old conversations.
- Built a pure-code **emotion engine** (no LLM in the hot path) grounded in attachment theory, simulating **three hormone axes with distinct half-lives** and an affection model spanning 8 relationship tiers, so emotional state persists and decays realistically across sessions.
- Implemented a bounded **agentic tool loop** (memory search/grep, timers) with offline “**dream**” **consolidation** (clustering, LLM naming, centroid refresh) and a nightly distillation agent.
- Guaranteed correctness with **144 offline unit tests** (<1s) and **PostgreSQL advisory locks** for cross-process safety; now re-architecting the state machine into **decoupled, agent-mediated tools** (v0.7).

Onyx: AI-Enhanced Personal Analytics Platform | *Python, Flask, PostgreSQL, Redis* Oct. 2025 – Present

- Engineered a containerized full-stack platform serving **RESTful APIs** on AWS EC2 with a **Flask/Gunicorn** backend (Gevent non-blocking async I/O) behind an **Nginx** reverse proxy with SSL termination and multi-worker fault tolerance.
- Achieved **~150ms** end-to-end latency for multi-device text synchronization using **Server-Sent Events (SSE)** and **Redis Pub/Sub**, ensuring atomic state updates across distributed backend workers.
- Built a **RAG** pipeline over user feedback and activity logs (**text-embedding-3-small** + pgvector cosine search), driving a **34%** improvement in suggestion accuracy; visualized results in **React** dashboards.

Aetheris: High-Performance RL Engine for Snake | *Rust, PyTorch, Rayon, PyO3* Dec. 2025 – Mar. 2026

- Built an RL simulator in **Rust** with **Rayon** (128 parallel environments) and **PyO3** zero-copy exchange to bypass the Python GIL, cutting cross-language overhead by **45%**; a **BFS floodfill** heuristic accelerated convergence by **300%**.
- Implemented **Double Dueling DQN** with **Prioritized Experience Replay**, stabilizing Q-value estimates and eliminating model collapse; achieved a peak score of **162** (average **105**) on a 20×20 grid.

EXPERIENCE

Math Club President

Bayview Secondary School

- Mentored 30+ students in advanced algorithms and competition math, leading the team to **13th place globally** in HMMT Team Guts.

Richmond Hill, ON

Sept. 2024 – Sept. 2025